Lie's Theorem for Solvable Complex Lie Algebras

An End-to-End Meta-Math Walkthrough

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Abstract

An ideal of codimension one provides a weight vector by induction. A trace calculation prevents the remaining direction from changing that weight, enabling one last eigenvector selection. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

RT-04 / Representation Theory and Lie Theory / Report v1

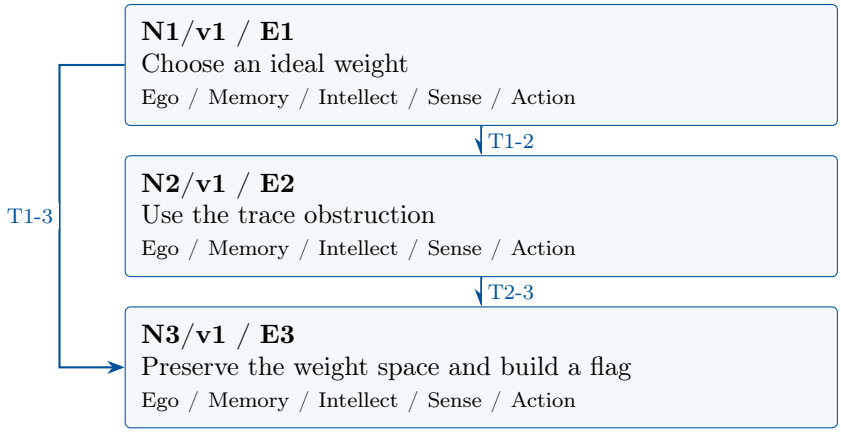
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $\mathfrak{g}$ be a finite-dimensional solvable complex Lie algebra and $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ a representation on a nonzero finite-dimensional complex vector space. There is a common eigenvector: some $v \neq 0$ and linear $\lambda : \mathfrak{g} \to \mathbb{C}$ satisfy $\rho(y)v = \lambda(y)v$ for all y. Consequently there is a basis in which every $\rho(y)$ is upper triangular. Solvable means the iterated derived series eventually becomes zero.*

A. Foundations and permitted imports

Use eigenvalue existence over $\mathbb{C}$, trace of a commutator equal to zero, induction on dimension, and the representation identity $\rho([x, y]) = [\rho(x), \rho(y)]$. Write yv for $\rho(y)v$. A subalgebra of a solvable algebra is solvable. A nonzero solvable algebra has $[\mathfrak{g}, \mathfrak{g}] \neq \mathfrak{g}$, since equality would make its derived series constant and nonzero.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. There exist x with $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{C}x$, a nonzero v_0, and a linear weight λ on $\mathfrak{h}$ with $yv_0 = \lambda(y)v_0$.

Receiving activity. N2.1

Compatibility. E1/v1 and $W = \text{span}\{v_0, xv_0, x^2v_0, \ldots\}$. Check: The finite span W is invariant under both x and $\mathfrak{h}$, so restricted commutator traces are legitimate.

T1-3: N1/E1/v1 to N3. There exist x with $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{C}x$, a nonzero v_0, and a linear weight λ on $\mathfrak{h}$ with $yv_0 = \lambda(y)v_0$.

Receiving activity. N3.1

Compatibility. E1/v1 and E2/v1; $E = \{v : yv = \lambda(y)v \text{ for all } y \in \mathfrak{h}\}$. Check: $E \neq 0$ since it contains v_0; the final induction decreases representation dimension.

T2-3: N2/E2/v1 to N3. $\lambda([x, y]) = 0$ for every $y \in \mathfrak{h}$.

Receiving activity. N3.1

Compatibility. E1/v1 and E2/v1; $E = \{v : yv = \lambda(y)v \text{ for all } y \in \mathfrak{h}\}$. Check: $E \neq 0$ since it contains v_0; the final induction decreases representation dimension.

III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

A. N1/v1: Choose an ideal weight

Object and role. Reduce the Lie algebra dimension, then record the orbit of a weight vector under the missing direction. **Local obligation:** There exist x with $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{C}x$, a nonzero v_0, and a linear weight λ on $\mathfrak{h}$ with $yv_0 = \lambda(y)v_0$.

Inputs. Induction on $d = \dim \mathfrak{g}$ and a hyperplane $\mathfrak{h} \supset [\mathfrak{g}, \mathfrak{g}]$.

E — Ego

Publish E1/v1 only after establishing this obligation: There exist x with $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{C}x$, a nonzero v_0, and a linear weight λ on $\mathfrak{h}$ with $yv_0 = \lambda(y)v_0$.

M — Memory

Induction on $d = \dim \mathfrak{g}$ and a hyperplane $\mathfrak{h} \supset [\mathfrak{g}, \mathfrak{g}]$. Retain the stated assumptions and import versions.

I — Intellect

Reduce the Lie algebra dimension, then record the orbit of a weight vector under the missing direction.

S — Sense

$\mathfrak{h}$ is an ideal because every bracket with it lies in $[\mathfrak{g}, \mathfrak{g}]$; induction decreases algebra dimension.

A — Action

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 When $\mathfrak{g} = 0$, any nonzero vector works. For $d > 0$, solvability makes the derived algebra proper. Choose a hyperplane $\mathfrak{h}$ containing it and x outside. The hyperplane is an ideal and solvable, of dimension $d - 1$.

N1.2 Apply the induction hypothesis to the restriction of the representation to $\mathfrak{h}$. It gives $v_0 \neq 0$ with $yv_0 = \lambda(y)v_0$. The scalar function λ is linear because the representation is linear and v_0 is nonzero.

Export E1/v1. There exist x with $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{C}x$, a nonzero v_0, and a linear weight λ on $\mathfrak{h}$ with $yv_0 = \lambda(y)v_0$.

Acceptance gate. $\mathfrak{h}$ is an ideal because every bracket with it lies in $[\mathfrak{g}, \mathfrak{g}]$; induction decreases algebra dimension.

Targeted repair. If an arbitrary hyperplane is assumed ideal, choose one containing the derived algebra before applying the commutator calculation. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Use the trace obstruction

Object and role. Triangularize the ideal action on a finite cyclic span and take traces of commutators there. **Local obligation:** $\lambda([x, y]) = 0$ for every $y \in \mathfrak{h}$.

Inputs. E1/v1 and $W = \operatorname{span}\{v_0, xv_0, x^2v_0, \ldots\}$. Incoming proof exports: E1/v1 at N2.1.

E — Ego

Publish E2/v1 only after establishing this obligation: $\lambda([x, y]) = 0$ for every $y \in \mathfrak{h}$.

M — Memory

E1/v1 and $W = \operatorname{span}\{v_0, xv_0, x^2v_0, \ldots\}$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

I — Intellect

Triangularize the ideal action on a finite cyclic span and take traces of commutators there.

S — Sense

The finite span W is invariant under both x and $\mathfrak{h}$, so restricted commutator traces are legitimate.

A — Action

Carry out activities N2.1–N2.3 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Put $v_j = x^j v_0$. Induction on j, simultaneously for every $y \in \mathfrak{h}$, gives $yv_j - \lambda(y)v_j \in \operatorname{span}(v_0, \ldots, v_{j-1})$. Indeed $yxv_{j-1} = x(yv_{j-1}) + [y, x]v_{j-1}$; the first term has the required leading part and the second lies in the previous span because $[y, x] \in \mathfrak{h}$ and the induction hypothesis applies to it.

N2.2 Let $m \geq 1$ be the first index where v_m depends on its predecessors. It exists by finite dimension. Then $v_0, \ldots, v_{m-1}$ form a basis of W, which is x-invariant and, by the preceding formula, $\mathfrak{h}$-invariant. For $y \in \mathfrak{h}$, its action on W is triangular with all diagonal entries $\lambda(y)$.

N2.3 The action of $[x, y]$ on W is the commutator of the two restricted operators, so its trace is zero. It also has trace $m\lambda([x, y])$, since $[x, y] \in \mathfrak{h}$. Characteristic zero gives $\lambda([x, y]) = 0$.

Export E2/v1. $\lambda([x, y]) = 0$ for every $y \in \mathfrak{h}$.

Acceptance gate. The finite span W is invariant under both x and $\mathfrak{h}$, so restricted commutator traces are legitimate.

Targeted repair. If a trace is taken on a noninvariant partial span, first pass to the full finite cyclic space W and establish both invariances. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Preserve the weight space and build a flag

Object and role. Use the vanished commutator weights to show $xE \subset E$, then repeat on quotient spaces. **Local obligation:** A common eigenvector exists, and the representation admits a full invariant flag and upper-triangular matrices.

Inputs. E1/v1 and E2/v1; $E = \{v : yv = \lambda(y)v \text{ for all } y \in \mathfrak{h}\}$. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1.

E — Ego

Publish E3/v1 only after establishing this obligation: A common eigenvector exists, and the representation admits a full invariant flag and upper-triangular matrices.

M — Memory

E1/v1 and E2/v1; $E = \{v : yv = \lambda(y)v \text{ for all } y \in \mathfrak{h}\}$. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1. Retain the stated assumptions and import versions.

I — Intellect

Use the vanished commutator weights to show $xE \subset E$, then repeat on quotient spaces.

S — Sense

$E \neq 0$ since it contains v_0; the final induction decreases representation dimension.

A — Action

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 For $v \in E$ and $y \in \mathfrak{h}$, $y(xv) = x(yv) + [y, x]v = \lambda(y)xv + \lambda([y, x])v = \lambda(y)xv$. Thus $xE \subset E$. On this nonzero finite-dimensional complex space, x has an eigenvector v. Since $\mathfrak{g} = \mathfrak{h} + \mathbb{C}x$, it is a common eigenvector for all of $\mathfrak{g}$, completing the induction on d.

N3.2 The line $\mathbb{C}v$ is invariant. To triangularize, induct separately on $\dim V$: the quotient $V/\mathbb{C}v$ has a representation of the same solvable algebra and a full invariant flag by induction. Lift its flag and prepend $\mathbb{C}v$. A basis adapted to this flag makes all representing matrices upper triangular.

Export E3/v1. A common eigenvector exists, and the representation admits a full invariant flag and upper-triangular matrices.

Acceptance gate. $E \neq 0$ since it contains v_0; the final induction decreases representation dimension.

Targeted repair. If E is claimed invariant without E2, restore the commutator term and use its zero weight; otherwise the eigenvector argument is unsupported. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Choose an ideal weight

1. **N1.1.** When $\mathfrak{g} = 0$, any nonzero vector works. For $d > 0$, solvability makes the derived algebra proper. Choose a hyperplane $\mathfrak{h}$ containing it and x outside. The hyperplane is an ideal and solvable, of dimension $d - 1$.

2. **N1.2.** Apply the induction hypothesis to the restriction of the representation to $\mathfrak{h}$. It gives $v_0 \neq 0$ with $yv_0 = \lambda(y)v_0$. The scalar function λ is linear because the representation is linear and v_0 is nonzero.

B. N2 — Use the trace obstruction

1. **N2.1.** Put $v_j = x^j v_0$. Induction on j, simultaneously for every $y \in \mathfrak{h}$, gives $yv_j - \lambda(y)v_j \in \operatorname{span}(v_0, \ldots, v_{j-1})$. Indeed $yxv_{j-1} = x(yv_{j-1}) + [y, x]v_{j-1}$; the first term has the required leading part and the second lies in the previous span because $[y, x] \in \mathfrak{h}$ and the induction hypothesis applies to it.

2. **N2.2.** Let $m \geq 1$ be the first index where v_m depends on its predecessors. It exists by finite dimension. Then $v_0, \ldots, v_{m-1}$ form a basis of W, which is x-invariant and, by the preceding formula, $\mathfrak{h}$-invariant. For $y \in \mathfrak{h}$, its action on W is triangular with all diagonal entries $\lambda(y)$.

3. **N2.3.** The action of $[x, y]$ on W is the commutator of the two restricted operators, so its trace is zero. It also has trace $m\lambda([x, y])$, since $[x, y] \in \mathfrak{h}$. Characteristic zero gives $\lambda([x, y]) = 0$.

C. N3 — Preserve the weight space and build a flag

1. **N3.1.** For $v \in E$ and $y \in \mathfrak{h}$, $y(xv) = x(yv) + [y, x]v = \lambda(y)xv + \lambda([y, x])v = \lambda(y)xv$. Thus $xE \subset E$. On this nonzero finite-dimensional complex space, x has an eigenvector v. Since $\mathfrak{g} = \mathfrak{h} + \mathbb{C}x$, it is a common eigenvector for all of $\mathfrak{g}$, completing the induction on d.

2. **N3.2.** The line $\mathbb{C}v$ is invariant. To triangularize, induct separately on $\dim V$: the quotient $V/\mathbb{C}v$ has a representation of the same solvable algebra and a full invariant flag by induction. Lift its flag and prepend $\mathbb{C}v$. A basis adapted to this flag makes all representing matrices upper triangular.

□

V. WORKED ORCHESTRATION

Let $\mathfrak{g}$ be the upper-triangular 2×2 complex matrices acting on $\mathbb{C}^2$. Its derived algebra consists of strictly upper-triangular matrices, whose mutual brackets vanish, so it is solvable. The vector e_1 is a common eigenvector with weight $\lambda\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) = a$. The invariant flag is $0 \subset \mathbb{C}e_1 \subset \mathbb{C}^2$.

A. Boundary and hypothesis audit

The standard representation of $\mathfrak{sl}_2(\mathbb{C})$ has no common eigenvector: the raising matrix kills only $\mathbb{C}e_1$, whereas the lowering matrix sends e_1 to e_2. This Lie algebra is not solvable. Over $\mathbb{R}$, even a one-dimensional abelian algebra can act by a rotation without a real eigenvector.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N3 step: the $\mathfrak{h}$-weight space is automatically invariant under x. The identity $y(xv) = \lambda(y)xv + \lambda([y,x])v$ exposes the missing term. **Diagnosis:** Intellect omitted the trace obstruction. **Repair:** establish N2 on the invariant cyclic span, then substitute $\lambda([y,x]) = 0$. This fills a genuine logical gap under the original hypotheses; recheck the common eigenvector and flag exports.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Compare this invariant-flag theorem with the highest-weight structure of nonsolvable $\mathfrak{sl}_2$ in RT-05.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Etingof and collaborators, Introduction to Representation Theory, sections on representations, characters, and $\mathfrak{sl}_2$; MIT Lie Groups and Lie Algebras I lecture notes for solvable Lie algebras.